Evaluation of Measurement Uncertainty

[In accordance with ISO Guide 98-3: 2008]
DETERMINATION OF POINT LOAD STRENGTH INDEX OF ROCK
ASTM D5731-16

1 Test equipment
Point Load Tester
Steel Rule / Measuring tape
Caliper

2 Mathematical model
$I_{S(50)} = F * I_S$
and
$I_S = P / D_e^2$; $F = (D_e / 50)^{0.45}$
and
$D_e^2 = D^2$ (for diametral tests without penetration) or $D_e^2 = D \times D'$ (for diametral tests with penetration) or
$D_e^2 = 4WD / \pi$ (for diametral tests without penetration) or $D_e^2 = 4WD' / \pi$ (for diametral tests with penetration)

where $I_{S(50)}$ = size corrected point load index in MPa
F = size correction factor $(D_e / 50)^{0.45}$
I_S = uncorrected point load strength index in MPa
P = failure load in N
D_e = equivalent core diameter in mm
D = distance between platen contact point in mm
D' = distance between platen contact point measured at the instant of failure in mm
W = specimen width perpendicular to the loading direction in mm

3 Uncertainty equation
The contributory variances are as follows:
$I_{S(50)} = F * I_S$
$u_c^2(I_{S(50)}) = [c(F) \cdot u(F)]^2 + [c(I_S) \cdot u(I_S)]^2$

with the sensitivity coefficient:
$c(F) = \partial I_{S(50)} / \partial F = I_S$
$c(I_S) = \partial I_{S(50)} / \partial I_S = F$

$I_S = P / D_e^2$; $F = (D_e / 50)^{0.45}$
$u^2(I_S) = [c(P) \bullet u(P)]^2 + [c(D_e) \bullet u(D_e)]^2$ $u^2(F) = [c(D_e) \bullet u(D_e)]^2$

with the sensitivity coefficient: with the sensitivity coefficient:
$c(P) = \partial I_S / \partial P = 1 / D_e^2$ $c(D_e) = \partial F / \partial D_e = 0.45(D_e)^{(-0.55)} / (50^{0.45})$
$c(D_e) = \partial I_S / \partial D_e = -2P / D_e^3$

D_e = D (for diametral tests without penetration) A or D_e = (D x D')^1/2 (for diametral tests with penetration) B or
$u^2(D_e) = [c(D) \bullet u(D)]^2$ $u^2(D_e) = [c(D) \bullet u(D)]^2 + [c(D') \bullet u(D')]^2$

with the sensitivity coefficient: with the sensitivity coefficient:
$c(D) = \partial D_e / \partial D = 1$ $c(D) = \partial D_e / \partial D = D'/[2*(D*D')^{1/2}]$
$c(D') = \partial D_e / \partial D' = D/[2*(D*D')^{1/2}]$

D_e = (4WD' / π)^1/2 (for axial tests without penetration) C or De = (4WD' / π)^1/2 (for axial tests with penetration) D
$u^2(D_e) = [c(W) \bullet u(W)]^2 + [c(D) \bullet u(D)]^2$ $u^2(D_e) = [c(W) \bullet u(W)]^2 + [c(D') \bullet u(D')]^2$

with the sensitivity coefficient: with the sensitivity coefficient:
$c(W) = \partial D_e / \partial W = D/(WD\pi)^{1/2}$ $c(W) = \partial D_e / \partial W = D'/(WD'\pi)^{1/2}$
$c(D) = \partial D_e / \partial D = W/(WD/\pi)^{1/2}$ $c(D') = \partial D_e / \partial D' = W/(WD'\pi)^{1/2}$

4 Source of uncertainty
u(W) : uncertainty of the measured width perpendicular to load direction by steel rule / measuring tape consisting of:- (for axial tests with or without penetration)
- calibration uncertainty of the steel rule / measuring tape, $u(W_1)$
- uncertainty due to resolution of the steel rule / measuring tape, $u(W_2)$
- uncertainty due to error of the steel rule / measuring tape, $u(W_3)$
- uncertainty due to misalignment, $u(W_4)$

u(D) : uncertainty of the measured dimension by caliper consisting of:- (for diametral or axial tests without penetration)
- calibration uncertainty of the caliper, $u(D_1)$
- uncertainty due to resolution of the caliper, $u(D_2)$
- uncertainty due to error of the caliper, $u(D_3)$
- uncertainty due to misalignment, $u(D_4)$

u(D') : uncertainty of the measured distance between platens by steel rule consisting of:- (for diametral or axial tests with penetration)
- calibration uncertainty of the steel rule, $u(D'_1)$
- uncertainty due to resolution of the steel rule, $u(D'_2)$
- uncertainty due to error of the steel rule, $u(D'_3)$
- uncertainty due to misalignment, $u(D'_4)$

u(P) : uncertainty of the measured failure load by point load tester consisting of:-
- calibration uncertainty of the point load tester, $u(P_1)$
- uncertainty due to resolution(including fluctuation) of the point load tester, $u(P_2)$
- uncertainty due to error of the point load tester, $u(P_3)$

$u^2(W) = u^2(W_1) + u^2(W_2) + u^2(W_3)$ (3 measurements)

$u^2(D) = u^2(D_1) + u^2(D_2) + u^2(D_3)$ (3 measurements)

$u^2(D') = u^2(D) + u^2(D'_1) + u^2(D'_2) + u^2(D'_3)$

$u^2(P) = u^2(P_1) + u^2(P_2) + u^2(P_3)$

$u^2(I_S) = [c(P) \bullet u(P)]^2 + [c(D_e) \bullet u(D_e)]^2$

$u^2(F) = [c(D_e) \bullet u(D_e)]^2$

$u_c^2(I_{S(50)}) = [c(F) \cdot u(F)]^2 + [c(I_S) \cdot u(I_S)]^2$

6 Expanded uncertainty
Expanded uncertainty, $U_{IS} = k \bullet u_c[I_{S(50)}]$
where k is the coverage factor from Students' 't' table at a confidence level of approximately 95%.

7 **Input data**

u(W) **Caliper ID:**

Calibration uncertainty (mm)		(k =)
Resolution (mm)		
Error (mm)		
misalignment (mm)		

$u(W_1)$ (mm)	$u(W_2)$ (mm)	$u(W_3)$ (mm)	$u(W_4)$ (mm)	u(W) (mm)
#DIV/0!	0.0000	0.0000	0.0000	#DIV/0!

u(D) **Caliper ID:**

Calibration uncertainty (mm)		(k =)
Resolution (mm)		
Error (mm)		
misalignment (mm)		

$u(D_1)$ (mm)	$u(D_2)$ (mm)	$u(D_3)$ (mm)	$u(D_4)$ (mm)	u(D) (mm)
#DIV/0!	0.0000	0.0000	0.0000	#DIV/0!

u(D') **Caliper ID:**

Calibration uncertainty (mm)		(k =)
Resolution (mm)		
Error (mm)		
misalignment (mm)		

$u(D'_1)$ (mm)	$u(D'_2)$ (mm)	$u(D'_3)$ (mm)	$u(D'_4)$ (mm)	u(D') (mm)
#DIV/0!	0.0000	0.0000	0.0000	#DIV/0!

u(P) **Caliper ID:**

Calibration uncertainty (kN)		(k =) (U% =)
Resolution (kN)		
Error (kN)		(Error%)

$u(P_1)$ (N)	$u(P_2)$ (N)	$u(P_3)$ (N)	u(P) (N)
#DIV/0!	0.0000	0.0000	#DIV/0!

$u(D_e)$(for diametral tests without penetration)

u(D) (mm)	c(D)	$u(D_e)$ (mm)

$u(D_e)$(for diametral tests with penetration)

u(D) (mm)	c(D)	u(D') (mm)	c(D')	$u(D_e)$ (mm)

$u(D_e)$(for axial tests without penetration)

u(W) (mm)	c(W)	u(D) (mm)	c(D)	$u(D_e)$ (mm)

$u(D_e)$(for axial tests with penetration)

u(W) (mm)	c(W)	u(D') (mm)	c(D')	$u(D_e)$ (mm)

$u(I_S)$

u(P) (N)	c(P) (mm^{-2})	$u(D_e)$ (mm)	$c(D_e)$ (N/mm^3)	u(var)# (N/mm^3)	$u(I_S)$ (N/mm^2)

u(F)

$u(D_e)$	$c(D_e)$	u(F)

$u(I_{S(50)})$

u(F)	c(F) (MPa)	$u(I_S)$ (MPa)	$c(I_S)$	$u(I_{S(50)})$ (MPa)	$U(I_{S(50)})$ (MPa)

(k=2, CI=95%)